Aria Koul

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Pleasanton, CA

EDUCATION

University of California, Berkeley | Starting 2026
M.S. in Information and Data Science

University of California, Santa Cruz | Class of 2024
B.S. in Physics (Astrophysics), Minor in Statistics
GPA: 3.72

Selected Coursework: Introduction to Scientific Computing, Electricity/Magnetism/Optics, Physics and Machine Learning, Beginning Programming in Python, Mathematical Methods in Physics, Astrophysics Advanced Laboratory, Modern Physics, Scientific Communication and Writing

WORK EXPERIENCE

Picarro | Engineering Intern | April 2025 - July 2025

- Performed data analysis on experimental prototype data using Python to test the stability of a new optical feedback spectroscopy instrument.
- Tested spectroscopy analyzer data and debugged coding errors.

Airlinq | AI Engineering Intern | Dec 2024 - March 2025

- Created an AI agent using Python for customers to chat with regarding launching connected car services.
- Used Flask, LangChain, pandas, datetime, dotenv, and logging.

UC Santa Cruz | Research Assistant | July 2023 - Present

- Implemented a program to model cosmic rays' trajectories in the magnetosphere.
- Translated multiple physics algorithms and equations into Python.
- Used pandas, numpy, matplotlib, and scipy to analyze datasets with over 100,000 entries.

PROJECTS

Classifying Cosmic Rays | https://github.com/AriaKoul/cosmic-rays

- Developed a machine learning algorithm using neural networks to predict the type of particle in a primary cosmic ray, achieving a 90% accuracy rate and boosting research on cosmic rays.
- Optimized hyperparameters to reduce training time by 50%.
- Used pytorch, numpy, scikit-learn, matplotlib, and seaborn to analyze datasets with over 40,000 entries.

Stock Market Analysis | https://github.com/AriaKoul/stock-market-analysis

- Implemented a stock performance evaluation tool leveraging overnight and day gains of any stock over any time period, resulting in optimized investment strategies.
- Created a suite of unit tests to ensure reliability.
- Used pytest, pandas, numpy, matplotlib, and yfinance to analyze thousands of rows of data.

Memory Game | https://github.com/AriaKoul/memory_game | https://ariakoul.github.io/memory_game/

- Created a card matching game that includes easy, medium, and hard levels.
- Used HTML, CSS, and JavaScript to design the game.

DISTINCTIONS

Phi Beta Kappa Academic Honor Society, Member
A Counter Space Club, Treasurer

SKILLS

Python, HTML, CSS, Java, JavaScript, SQL, MATLAB, LaTeX